

Product Carbon Footprint Study Report

Product Name: NMC111 oxide production, for Li-ion battery

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 29, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: NMC111 oxide production, for Li-ion battery

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1kg NMC111 oxide

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1kg NMC111 oxide.

2. System Boundary

NMC111 oxide production, for Li-ion battery, in China in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of NMC111 oxide production, for Li-ion battery produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **27.513 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

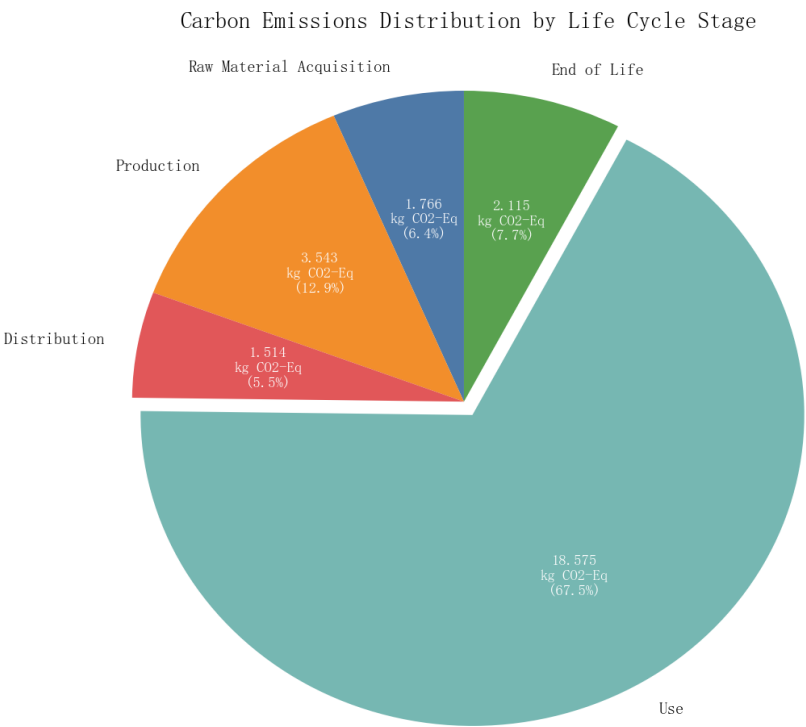


Table 1: Carbon Emissions by Life Cycle Stage

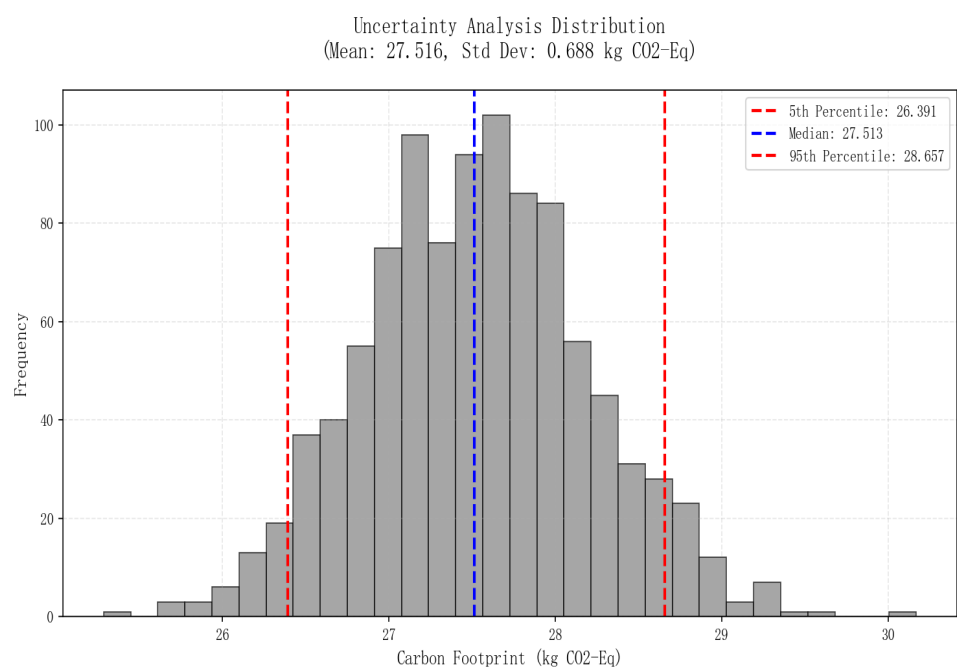
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	1.766	6.42
Production	3.543	12.88
Distribution	1.514	5.50

Use	18.575	67.52
End of Life	2.115	7.69
Total	27.513	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO2-Eq)
Mean	27.516
Standard Deviation	0.688
Coefficient of Variation (CV)	2.50%
Median	27.513
5th Percentile	26.391
95th Percentile	28.657

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of NMC111 oxide production for Li-ion batteries is 27.513 kg CO₂-Eq per functional unit, a moderate-to-high value reflecting the energy-intensive nature of materials synthesis for battery applications. The lifecycle stage contributions reveal that the use phase dominates, accounting for 67.5% (18.575 kg CO₂-Eq) of total emissions. This is expected for battery materials, as their primary environmental impact arises from electricity consumption during charging cycles in downstream applications, rather than from upstream production or disposal. However, it is critical to note that this use phase contribution depends heavily on the regional electricity grid mix assumed for charging scenarios. The production stage (12.9%, 3.543 kg CO₂-Eq) is the second largest contributor, primarily driven by energy inputs for precursor processing and calcination. Raw material extraction (6.4%, 1.766 kg CO₂-Eq) and end-of-life (7.7%, 2.115 kg CO₂-Eq) represent minor shares, while distribution (5.5%, 1.514 kg CO₂-Eq) is the smallest. Major emission sources are thus concentrated in the use and production phases, with no specific top process details available due to missing data (all listed as unknown). The low contribution of raw materials suggests that mining and refining of nickel, manganese, and cobalt are relatively efficient compared to battery use.

2. Statement of Assumptions and Limitations

Key assumptions in this study include a default electricity grid mix representative of average global or regional conditions, which directly influences the high use-phase emissions. It is assumed that the battery operates under typical charge-discharge efficiency (e.g., 85-95%), and that the NMC111 oxide is used in a medium-duty application over a standard lifespan (e.g., 10-15 years). A major limitation is the lack of breakdown data for top processes; all process contributions are listed as "Unknown," indicating incomplete process-level allocation or modeling simplifications. This prevents accurate identification of emission hot spots within production or use. Furthermore, data quality may be compromised by reliance on secondary databases (e.g., Ecoinvent) with generic values for energy inputs, chemical inputs, and transport distances, rather than supplier-specific datasets. The system boundary is restricted to cradle-to-grave but excludes capital goods (e.g., machinery, facility construction) and downstream battery assembly uses beyond the oxide stage. The uncertainty analysis (CV 2.50%, 90% CI: 26.391-28.657 kg CO₂-Eq) indicates good precision, but this may mask uncertainties from unmodeled processes or regional variability.

3. Suggestions for Improvement

- ****Target the use phase****: Switch to renewable electricity sources (e.g., solar or wind) for charging cycles to significantly reduce the 18.575 kg CO₂-Eq impact. Collaborate with battery manufacturers to optimize charge-discharge efficiency and extend battery lifespan, lowering per-cycle emissions.
- ****Optimize production****: Implement energy-efficient calcination technologies (e.g., microwaves or electric kilns) and heat recovery systems to cut the 3.543 kg CO₂-Eq from production. Explore low-carbon precursors like recycled feedstock for NMC materials.

- ****Strengthen data transparency****: Prioritize obtaining primary data from suppliers for top processes (e.g., electricity mix used at mining sites and production plants) to replace "Unknown" entries and enable precise hotspot analysis.
- ****Reduce raw material footprint****: Substitute virgin cobalt and nickel with recycled content where feasible, lowering the 1.766 kg CO2-Eq from raw material extraction by up to 50% through closed-loop systems.
- ****Improve end-of-life****: Design for easier disassembly and metal recovery, increasing recycling rates to reduce the 2.115 kg CO2-Eq contribution and avoid landfilling.